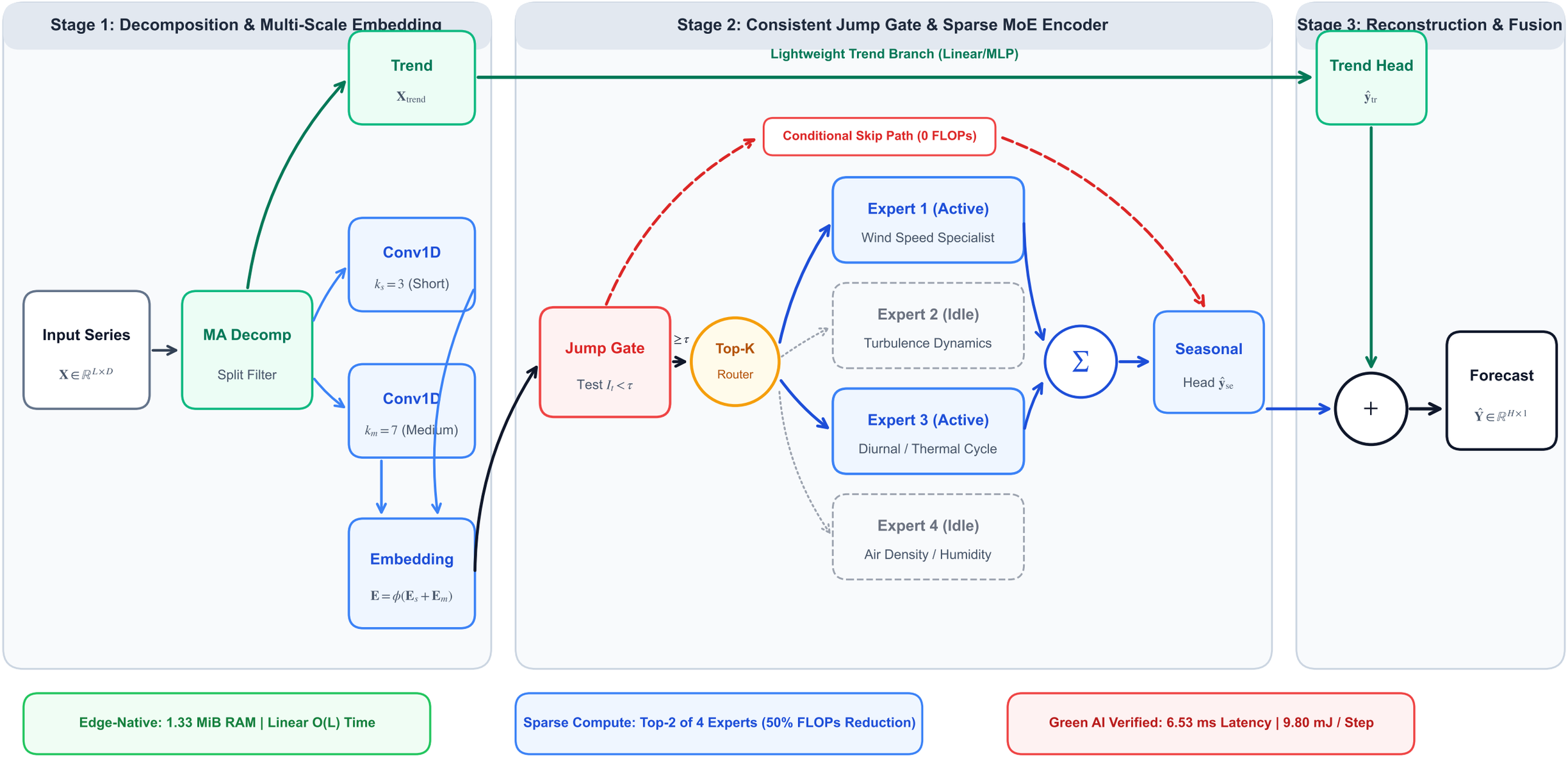

Stage 1: Decomposition & Multi-Scale Embedding
Stage 2: Consistent Jump Gate & Sparse MoE Encoder
Stage 3: Reconstruction & Fusion
Lightweight Trend Branch (Linear/MLP)
Input Series
$\mathbf{X} \in \mathbb{R}^{L \times D}$
MA Decomp
Split Filter
Trend
$\mathbf{X}_{\text{trend}}$
Conv1D
$k_s = 3$ (Short)
Conv1D
$k_m = 7$ (Medium)
Embedding
$\mathbf{E} = \phi(\mathbf{E}_s + \mathbf{E}_m)$
Jump Gate
Test $I_t < \tau$
$\geq \tau$
Top-K
Router
Conditional Skip Path (0 FLOPs)
Expert 1 (Active)
Wind Speed Specialist
Expert 2 (Idle)
Turbulence Dynamics
Expert 3 (Active)
Diurnal / Thermal Cycle
Expert 4 (Idle)
Air Density / Humidity
Σ
Seasonal
Head $\hat{\mathbf{y}}_{\text{se}}$
Trend Head
$\hat{\mathbf{y}}_{\text{tr}}$
+
Forecast
$\hat{\mathbf{Y}} \in \mathbb{R}^{H \times 1}$
Edge-Native: 1.33 MiB RAM | Linear O(L) Time
Sparse Compute: Top-2 of 4 Experts (50% FLOPs Reduction)
Green AI Verified: 6.53 ms Latency | 9.80 mJ / Step